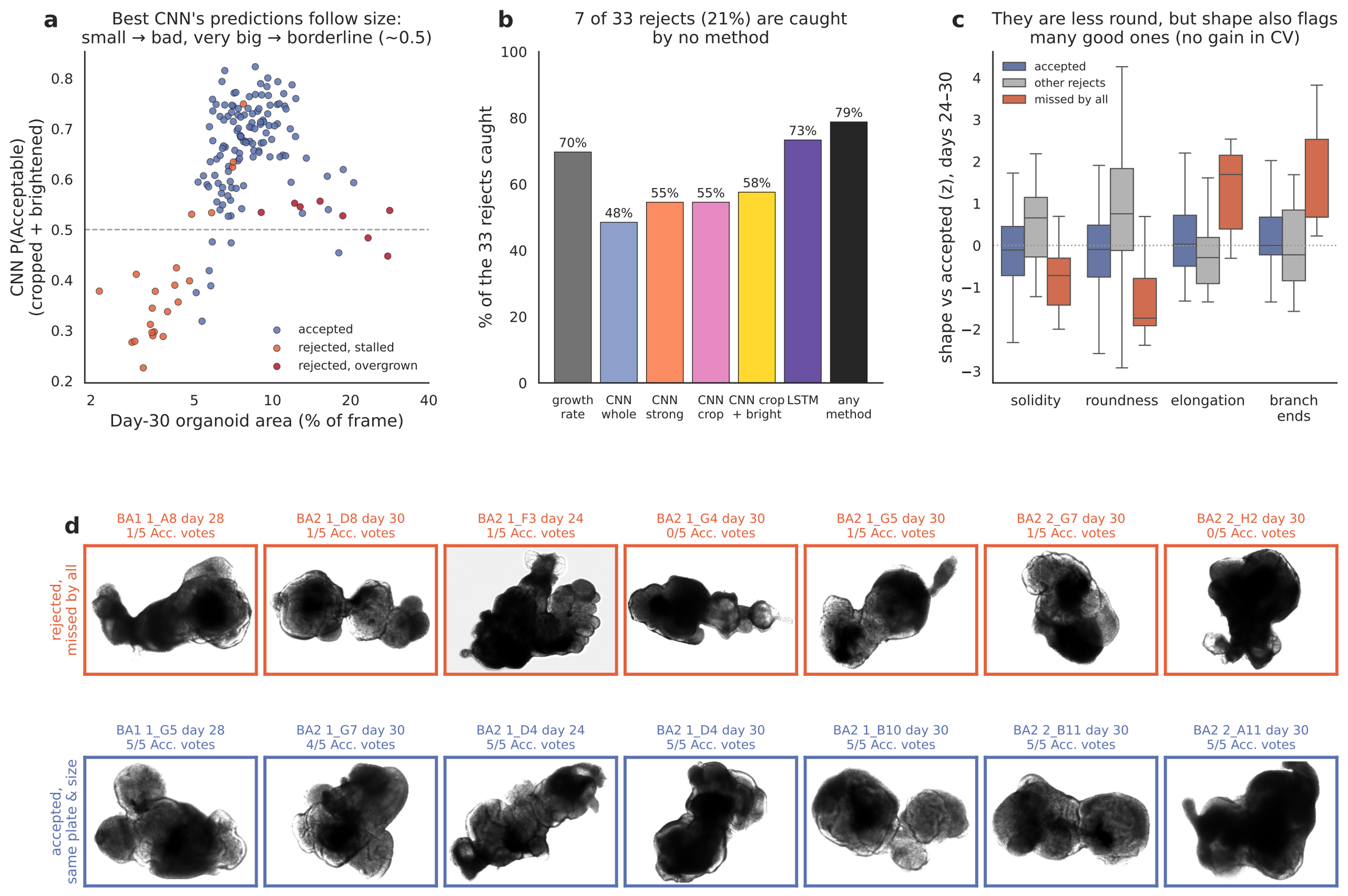

a
Best CNN's predictions follow size:
small → bad, very big → borderline (~0.5)
CNN P(Acceptable)
(cropped + brightened)
Day-30 organoid area (% of frame)
accepted
rejected, stalled
rejected, overgrown
b
7 of 33 rejects (21%) are caught
by no method
% of the 33 rejects caught
70%
48%
55%
55%
58%
73%
79%
growth rate
CNN whole
CNN strong
CNN crop
CNN crop + bright
LSTM
any method
c
They are less round, but shape also flags
many good ones (no gain in CV)
shape vs accepted (z), days 24–30
accepted
other rejects
missed by all
solidity
roundness
elongation
branch ends
d
rejected, missed by all
BA1 1_A8 day 28
1/5 Acc. votes
BA2 1_D8 day 30
1/5 Acc. votes
BA2 1_F3 day 24
1/5 Acc. votes
BA2 1_G4 day 30
0/5 Acc. votes
BA2 1_G5 day 30
1/5 Acc. votes
BA2 2_G7 day 30
1/5 Acc. votes
BA2 2_H2 day 30
0/5 Acc. votes
accepted, same plate & size
BA1 1_G5 day 28
5/5 Acc. votes
BA2 1_G7 day 30
4/5 Acc. votes
BA2 1_D4 day 24
5/5 Acc. votes
BA2 1_D4 day 30
5/5 Acc. votes
BA2 1_B10 day 30
5/5 Acc. votes
BA2 2_B11 day 30
5/5 Acc. votes
BA2 2_A11 day 30
5/5 Acc. votes